BlackSwanQuants Project

Intraday Momentum Strategy: Iterative Development and Systematic Improvement

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Abstract

This report documents the iterative development and backtesting of an intraday momentum strategy on QuantConnect, applied to the SPY ETF (tracking the S&P 500) over an 8-year period from May 2017 to May 2025, with an initial capital of $100,000. Starting from a base volatility-breakout model adapted from Zarattini et al. (2024), we systematically iterate through four refinements—decoupled exit frequency, EMA trend filtering, temporal persistence for exit confirmation, and a synthesis of the two most effective mechanisms—ultimately achieving a total return of 259%, a Sharpe Ratio of 1.036, a Sortino Ratio of 1.438, and a maximum drawdown of 8.4%. Each iteration is motivated by a precise diagnosis of the previous version's structural weaknesses. We document the strategy's relationship with its Buy-and-Hold benchmark. The report concludes with a structured agenda for further improvement, encompassing regime detection, multi-asset extension, and transaction cost optimisation.

Keywords: *Momentum* *Intraday Trading* *Algorithmic Trading* *QuantConnect* *SPY*

1. INTRODUCTION

The efficient market hypothesis claims that asset prices fully reflect all publicly available information, implying that consistently outperforming the market through active trading should not be feasible. However, decades of research have revealed persistent anomalies—among them the **momentum effect**. All variations of the momentum strategy rest on a simple assumption: assets that have performed well recently (winners) will continue to outperform assets that have performed poorly (losers) in the near future. The core of the strategy consists of buying past winners and selling past losers. Since its first formal documentation by Jegadeesh and Titman (1993), the momentum effect has been recorded across numerous asset classes and time horizons, suggesting it is a fundamental feature of financial market dynamics.

The work presented here builds directly upon Zarattini et al. (2024), who investigate the profitability of a simple intraday momentum strategy applied to SPY, one of the most liquid ETFs tracking the S&P 500. Unlike academic literature that typically limits trading to the last 30 minutes of the session, Zarattini et al. (2024) show that intraday trends can start earlier in the day, and demonstrate that a strategy using dynamic trailing stops anchored to time-varying noise boundaries achieves a Sharpe Ratio of 1.33 over the period 2007–2024 after transaction costs. Building upon this framework, our QuantConnect implementation incorporates several targeted modifications motivated by a structured diagnosis of each prior version's shortcomings.

The remainder of this report is structured as follows. Section 2 describes the baseline strategy. Section 3 presents baseline backtest results and a benchmark comparison. Section 4 provides the theoretical motivation for each iterative improvement. Section 5 presents the comparative performance of all five strategy versions. Section 6 outlines avenues for further development.

2. BASELINE STRATEGY

We implement an intraday momentum algorithm on QuantConnect using *historical volatility breakouts* anchored to each specific minute of the trading day. The design philosophy is intentionally parsimonious: the strategy seeks to exploit the persistence of intraday trends while strictly controlling overnight and regime risk through dynamic position sizing and a market-microstructure-aware composite exit rule.

2.1. Strategy Parameters

The baseline algorithm is initialised on QuantConnect with the core settings summarised in Table 1. The volatility target, lookback window, and leverage cap are the three free parameters of the model; all other settings are fixed by the trading environment or regulatory constraints.

Table 1. Baseline strategy configuration.

Parameter	Value	Note
Instrument	SPY	S&P 500 ETF
Bar resolution	1 minute	OHLCV data
Backtest period	May 2017 – May 2025	8 years
Initial capital	$100,000	
Vol. target	$\sigma_{\text{target}} = 2\%$	Daily
Lookback window	14 trading days	Rolling
Max leverage	2×	Hard cap
Entry cadence	Every 30 min	HH:00 and HH:30
Exit evaluation	Every 30 min	Composite rule
EOD liquidation	15:58 ET	Hard close
First tradeable bar	9:32 AM ET	Post-auction
Commission	$0.005/share	Min. $1/order

2.2. Signal Generation: Minute-Specific Volatility Bands

The central insight of the baseline signal is that intraday price diffusion is *not* uniform across the trading day. The well-documented U-shaped intraday volatility pattern means a fixed threshold would generate too many false signals near the open and too few near the close. Following Zarattini et al. (2024), we construct *minute-specific* breakout bands calibrated to the historical volatility at each clock time t.

Let t index a specific clock time within the trading day (e.g., 10:30 AM). The minute-specific noise level σ_t is the 14-day rolling mean of the absolute percentage deviation from the daily open at

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Backtested on QuantConnect. Strategy applied to SPY ETF, May 2017 – May 2025, initial capital $100,000. Built upon the volatility-breakout framework of Zarattini et al. (2024).

minute t:

$$\sigma_t = \frac{1}{14}\sum_{i=1}^{14}\left|\frac{P_{i,t}}{O_i} - 1\right| \tag{1}$$

where $P_{i,t}$ is the close price of the t-th minute bar on day i, and O_i is the opening price of day i. This quantity estimates the typical noise expected at clock time t given recent market conditions.

The dynamic breakout boundaries incorporate overnight gap information through reference price anchoring:

$$\mathrm{UB}_t = \max(O_{\text{today}}, C_{\text{yest}}) \times (1 + \sigma_t) \tag{2}$$

$$\mathrm{LB}_t = \min(O_{\text{today}}, C_{\text{yest}}) \times (1 - \sigma_t) \tag{3}$$

A price close above UB_t triggers a long signal; a close below LB_t triggers a short signal. By anchoring to max / min$(O_{\text{today}}, C_{\text{yest}})$, the bands automatically account for overnight gaps.

2.3. Dynamic Position Sizing

Position size is scaled inversely with recent realised volatility to maintain a constant risk profile across volatility regimes. Let σ_{daily} be the population standard deviation of daily close-to-close returns over the 14-day lookback. The target leverage factor is:

$$\Lambda = \min\left(2,\ \frac{\sigma_{\text{target}}}{\sigma_{\text{daily}}}\right) \tag{4}$$

The mechanism scales exposure upward during calm, low-volatility regimes (up to 1.5×) and automatically de-risks during turbulent ones. The hard cap is set at 2×, consistent with Zarattini et al, though backtesting revealed an artefact worth noting: uncapped leverage during low-volatility windows preceding large trending moves caused orders to exceed available free margin, triggering `Insufficient buying power` errors in the simulator. Rather than altering the parameter or introducing additional caps, we preserved the original specification to maintain coherence with the reference framework.

2.4. Composite Exit Rule

The exit rule is a composite trailing stop that triggers when price crosses *either* the active noise band boundary *or* the VWAP, whichever is reached first:

$$\text{Long exit if:}\quad P_t \le \max\big(\mathrm{UB}_t,\ \mathrm{VWAP}_t\big) \tag{5}$$

$$\text{Short exit if:}\quad P_t \ge \min\big(\mathrm{LB}_t,\ \mathrm{VWAP}_t\big) \tag{6}$$

The VWAP component is volume-adaptive, institutionally relevant, and self-widens throughout the session, giving winning trades room to run. VWAP is computed using market-hours data only, reanchored at each 9:30 AM open.

2.5. Execution Timing

Entry and exit conditions are evaluated every 30 minutes (HH:00 and HH:30) to reduce microstructure noise. The first bar of each trading day (9:30 AM) is reserved for recording the daily open price and is never used for signal generation; the earliest possible entry is therefore at the 9:32 AM bar evaluated at 10:00 AM.[1] All open positions are compulsorily liquidated at or after 15:58 ET to eliminate overnight gap risk.

2.6. Strategy Logic: Pseudocode

The following algorithm presents the complete event-driven logic of the base strategy as implemented on QuantConnect.

[1]The 30-minute cadence means the first evaluation after the open occurs at 10:00 AM; the 9:31 and 9:32 bars contribute to minute-specific statistics but do not trigger entries.

```
input: SPY minute bars, lookback L=14 days, vol_target=2%,
       max_lev=2, entry_interval=30 min

for each new minute bar do
    update sigma_t  <- 14-day rolling mean of |price/open - 1| at this
      ↪ clock time
    compute UB      <- max(open, prev_close) * (1 + sigma_t)
    compute LB      <- min(open, prev_close) * (1 - sigma_t)
    compute lev     <- min(max_lev, vol_target / sigma_daily)

    if time mod entry_interval == 0 and position != NONE then
        if (LONG  and price <= max(UB, VWAP))
           or (SHORT and price >= min(LB, VWAP)) then
            liquidate; reset position

    if time mod entry_interval == 0 and position == NONE then
        if   price > UB  then go LONG  with size = lev * equity / price
        elif price < LB  then go SHORT with size = lev * equity / price

    if time >= 15:58 ET then
        liquidate all
```

Code 1. Baseline Strategy – Minute-Specific Volatility Bands (pseudocode).

3. BASELINE BACKTEST RESULTS

Running the baseline strategy (Strategy 0) over the 8-year period yields stable, risk-averse results that serve as the reference for all subsequent improvements.

3.1. Key Performance Metrics

Table 2 reports the full set of performance statistics for Strategy 0 over May 2017 – May 2025.

Table 2. Strategy 0 (baseline) performance statistics, May 2017 – May 2025, initial capital $100,000.

Metric	Value	Comment
Total return	196%	On $100,000 initial capital
CAGR	14.5%	Annualised
Sharpe Ratio	0.835	Annualised, daily P&L
Sortino Ratio	1.075	Downside-deviation adjusted
Max drawdown	11.2%	Primary improvement target
Win rate	40%	
Loss rate	60%	
Avg. win	0.68%	Per trade
Avg. loss	−0.35%	Per trade
Profit-loss ratio	1.94	$\bar{w}/\lvert\bar{l}\rvert$
Expectancy	0.180	Normalised per trade
Total orders	3,594	
PSR	61.57%	Probabilistic Sharpe Ratio
Alpha	0.079	vs. SPY benchmark
Beta	−0.048	Near market-neutral

The Sharpe Ratio of 0.835 is a solid starting point for an intraday-only strategy, though it remains below the 1.0 threshold commonly regarded as the minimum for institutional deployment. The Sortino Ratio of 1.075 reveals that a significant share of total return volatility is upside: the strategy's profit-loss ratio of 1.94 and near-zero beta of −0.048 confirm that gains are driven by intraday trend capture rather than passive market exposure. The maximum drawdown of 11.2% is the primary target for improvement: the 30 minute exit evaluation window leaves the strategy exposed to sustained intraday reversals for up to 30 minutes between checks. The win rate of 40% is structurally coherent provided the average gain on winning trades substantially exceeds the average loss on losers, producing the positive P&L skew characteristic of the trend-following family; here the average win of 0.68% against an average loss of −0.35% satisfies this requirement with a profit-loss ratio of 1.94.

3.2. Comparison with the Buy-and-Hold Benchmark

During the entire 8-year period, the SPY benchmark delivered a total return of 166.4% , which Strategy 0 modestly exceeds with a total return of 196%. This absolute performance is notable for an intraday-only strategy that remains flat overnight, thereby intentionally sacrificing the well-documented overnight equity premium. The strategy's advantage becomes clearer on a *risk-adjusted basis*. The benchmark's annualized volatility historically stands at 16.0% , yielding a Buy-and-Hold Sharpe Ratio of 0.47 , whereas Strategy 0 achieves a nearly double Sharpe of 0.835. More importantly, the benchmark's maximum drawdown of 33.7% (observed during the March 2020 market crash) substantially exceeds Strategy 0's tightly controlled 11.2%.

Recent high-volatility window: Feb–Apr 2025. During the recent 3-month window, Strategy 0 delivered a net return of 4.70% with a Sharpe Ratio of 0.826, a Sortino Ratio of 1.422, and a maximum drawdown of only 5.4%. In contrast, the SPY sustained a maximum drawdown of 20.0% and a net loss of 5.49% driven by escalating macroeconomic uncertainty. Strategy 0's profit-loss ratio of 1.98 and a near-zero beta of 0.03 confirm that these gains were generated by pure intraday trend capture rather than passive market exposure. This behaviour directly aligns with the empirical findings of Zarattini et al. (2024), who document a monotonically increasing relationship between the VIX level at market open and the strategy's Sharpe Ratio. Elevated implied volatility expands the magnitude of intraday ranges. It is also associated with sustained institutional order-flow imbalances that momentum-based strategies are designed to exploit.

3.3. Structural Weaknesses of the Baseline

Two principal vulnerabilities emerge from the equity curve and trade log. First, the 30-minute exit window creates structural latency: once a trend reversal is underway, the strategy can suffer up to 30 minutes of adverse price action before re-evaluating the composite exit condition. Second, the strategy enters breakout trades irrespective of the broader intraday trend structure, generating false positives at its maximum rate during mean-reverting regimes. These two weaknesses motivate the iterative improvements described in the following section.

4. ITERATIVE IMPROVEMENTS: MOTIVATION AND DESIGN

The evolution from Strategy 0 to Strategy 4 can be understood as a systematic investigation of a single tension we term the **Momentum Paradox**: the same granularity of exit monitoring that protects capital during reversals also destroys capital through false exits during valid trend continuations. Each iteration navigates this trade-off from a different angle.

4.1. Strategy 1: Decoupled Exit Frequency

Motivation. The most direct fix to Strategy 0's slow exit resolution is to increase the exit-monitoring frequency while keeping entry evaluation at 30 minutes. This decouples two logically independent tasks: *entry*—a 30-minute decision about whether a breakout is statistically significant—and *risk management*—a higher-frequency check on whether the position remains structurally valid.

Design. Strategy 1 retains the 30-minute entry cadence but introduces a 5-minute exit-check interval ($\Delta t_{\text{exit}} = 5$ min). The composite exit condition (VWAP or band crossing) is re-evaluated every 5 minutes, reducing the maximum adverse excursion window from 30 minutes to 5 minutes. In the QuantConnect implementation, this is realised by evaluating the exit block whenever `current\_time.minute \% exit\_check\_interval == 0` with `exit\_check\_interval = 5`, while entry remains gated on `current\_time.minute \% entry\_check\_interval == 0` with `entry\_check\_interval = 30`.

Observed results. Table 3 records the key metrics for Strategy 1 alongside the Strategy 0 baseline.

Table 3. Strategy 1 vs. Strategy 0 performance metrics.

Metric	Strategy 0	Strategy 1
Total return (%)	196.00	177.00
Sharpe Ratio	0.835	0.84
Max drawdown (%)	11.2	10.1
Win rate (%)	40	32
Total orders	3,594	4,900
Exit check interval	30 min	5 min

Interpretation. Strategy 1 achieves a reduction in maximum drawdown (11.2% → 10.1%), confirming the benefit of faster risk monitoring. However, the win rate collapses from 40% to 32% and total return *falls* from 196% to 177%. The explanation is mechanical: VWAP is a slow-moving quantity that generates brief, transient violations of the composite exit condition with no predictive value for the ultimate direction of the trade. By checking every 5 minutes without any noise filter, Strategy 1 frequently closes valid positions on these transient violations—the classic whipsaw outcome. The lesson: monitoring speed alone is insufficient without a filter for statistical significance.

4.2. Strategy 2: EMA Trend Filter

Motivation. Rather than managing risk through faster exits, Strategy 2 addresses the root cause upstream: it filters out structurally low-probability trades at the point of entry.

Design. Strategy 2 reverts to symmetric 30-minute cadence for both entries and exits. It introduces a 100-period Exponential Moving Average (EMA) as a trend-confirmation filter on 1-minute bars, requiring dual confirmation for entry:

$$\text{Long entry:} \quad P_t > \text{UB}_t \ \text{ and } \ P_t > \text{EMA}_{100}(t) \tag{7}$$

$$\text{Short entry:} \quad P_t < \text{LB}_t \ \text{ and } \ P_t < \text{EMA}_{100}(t) \tag{8}$$

The 100-minute lookback captures approximately two hours of price history, bridging the gap between microstructure noise (resolved in the first 20–30 minutes) and the structural intraday trend (which typically persists for 2–4 hours). In the QuantConnect implementation, the EMA is instantiated as `self.ema = self.EMA(self.spy\_symbol, 100, Resolution.Minute)`.

Table 4. Strategy 2 vs. Strategy 0 and 1 performance metrics.

Metric	S0	S1	S2
Total return (%)	196.00	177.00	227.00
Sharpe Ratio	0.835	0.84	0.94
Max drawdown (%)	11.2	10.1	8.7
Win rate (%)	40	32	40
Total orders	3,594	4,900	3,450
EMA filter	No	No	Yes (100-period)

Observed results.

Interpretation. The EMA filter produces a meaningfully different trade set, reducing orders to 3,450 and recovering the win rate to 40%, with maximum drawdown improving to 8.7%—the best absolute drawdown of any strategy tested prior to Strategy 4—and total return rising to 227% with a Sharpe of 0.94. This result affirms a well-known principle in systematic trading: entry quality compounds multiplicatively,

while exit timing is additive. The residual limitation of Strategy 2 is its slow exit cadence: once a position is open, the composite exit condition is still evaluated only every 30 minutes, leaving capital exposed to protracted intraday reversals that a faster exit mechanism could contain.

4.3. Strategy 3: Temporal Persistence Framework

Motivation. Strategies 1 and 2 reveal complementary insights: Strategy 1 monitors exits frequently but indiscriminately, generating excessive false exits, while Strategy 2 enters trades selectively but monitors exits slowly, leaving residual exposure to late-stage reversals. The ideal combination is fast monitoring paired with a statistical noise filter on the exit signal itself. Strategy 3 implements this by requiring an exit signal to *persist* across multiple consecutive checks before acting.

Design. Strategy 3 reinstates the decoupled frequency structure from Strategy 1 (30-minute entries, 5-minute exit checks) and abandons the EMA filter. It introduces an exit confirmation counter: a detected violation is acted upon only if it persists for $N = 4$ consecutive 5-minute checks (i.e., 20 continuous minutes). In the QuantConnect implementation, this is realised through counters `self.long\_exit\_counter` and `self.short\_exit\_counter`, incremented each time the composite exit condition is breached and reset to zero when the condition is no longer satisfied. An exit is triggered only when the relevant counter reaches `self.exit\_confirmation\_bars = 4`.

Formally, an exit at time τ requires that for all $k \in \{0, 1, 2, 3\}$ the exit signal $s(\tau - k \cdot 5\,\text{min})$ is active.

Note

The 4-bar sequential validation functions as a causal low-pass filter applied to the binary exit signal, with an effective cutoff at periods shorter than 20 minutes. Noise components with periods below 20 minutes are attenuated, while genuine structural reversals (persisting ≥ 20 minutes) pass through unimpeded.

Table 5. Strategy 3 vs. all prior versions: full performance comparison.

Metric	S0	S1	S2	S3
Total return (%)	196.00	177.00	227.00	252.00
Sharpe Ratio	0.835	0.84	0.94	1.012
Max drawdown (%)	11.2	10.1	8.7	9.3
Win rate (%)	40	32	40	40
Total orders	3,594	4,900	3,450	3,550
Entry cadence	30 min	30 min	30 min	30 min
Exit cadence	30 min	5 min	30 min	5 min
EMA filter	No	No	Yes	No
Exit persistence bars	—	—	—	4

Observed results. Strategy 3 achieves the highest performance across all primary institutional metrics up to this point. Compared with the baseline, it improves the Sharpe from 0.835 to 1.012, raises total return from 196% to 252%, and cuts maximum drawdown from 11.2% to 9.3%. The crossing of the Sharpe 1.0 threshold is particularly meaningful: it signals that the strategy generates more than one unit of excess return per unit of annualised risk, the canonical minimum for institutional deployment. However, Strategy 3 does not include an entry-quality filter: it benefits from a noise-robust exit mechanism but remains exposed to directionally misaligned entries during mean-reverting regimes.

Figure 1. Standalone equity curve of Strategy 4.

4.4. Strategy 4: Combined EMA Filter and Temporal Persistence

Motivation. The progression from Strategy 0 to Strategy 3 reveals that entry quality and exit robustness address orthogonal sources of performance degradation. Strategy 2 demonstrates that restricting entry to EMA-confirmed trend regimes dramatically improves the signal-to-noise ratio of the trade set: fewer, higher-quality trades yield a superior risk-adjusted return. Strategy 3 demonstrates that wrapping the exit signal in a temporal persistence filter eliminates the whipsaw losses that plagued Strategy 1. The two mechanisms are structurally independent: the EMA filter operates at the entry decision gate, while the persistence counter operates on the exit signal after a position is open. Their combination is therefore not merely additive but potentially synergistic—higher-quality entries are held for longer against noise, and the persistence gate prevents premature liquidation of the premium trades that pass the EMA screen.

Formally, the hypothesis motivating Strategy 4 can be stated as follows:

Hypothesis 1. *Let Q_{entry} denote the distributional quality of the entry trade set (measured by expected P&L per trade conditional on direction) and Q_{exit} the statistical robustness of the exit trigger (measured by false-exit rate). If Q_{entry} and Q_{exit} are independently improvable, a mechanism that maximises both simultaneously dominates any mechanism that maximises only one.*

From a signal-processing perspective, the EMA filter acts as a *pre-filter* that selects only those breakout signals whose direction is confirmed by the slower-timescale trend, while the temporal persistence gate acts as a *post-filter* that suppresses transient reversions in the exit signal below the 20-minute persistence threshold. The composition of a pre-filter and a post-filter is a well-established technique in digital signal processing for improving the signal-to-noise ratio of binary detection systems—here applied to the stochastic binary signals arising from intraday price dynamics.

Design. Strategy 4 activates both mechanisms simultaneously. Entry requires a volatility breakout *and* EMA confirmation, identically to Strategy 2:

$$\text{Long entry:} \quad P_t > \text{UB}_t \ \text{ and } \ P_t > \text{EMA}_{100}(t) \tag{9}$$

$$\text{Short entry:} \quad P_t < \text{LB}_t \ \text{ and } \ P_t < \text{EMA}_{100}(t) \tag{10}$$

Exit monitoring is decoupled from entry at a 5-minute cadence, with the same 4-bar persistence gate as Strategy 3. An exit is triggered only when the composite condition (VWAP or band crossing) is satisfied for $N = 4$ consecutive 5-minute checks:

$$\text{Exit triggered} \iff \sum_{k=0}^{N-1} \mathbf{1}[s(\tau - k \cdot \Delta t) = 1] = N \tag{11}$$

where $\Delta t = 5$ min, $N = 4$, and $s(\cdot) \in \{0, 1\}$ is the binary exit signal at each check. The full parameter configuration is summarised in Table 6.

Table 6. Strategy 4 configuration, highlighting the two active mechanisms.

Component	Setting	Source
Entry cadence	30 min	All versions
Exit cadence	5 min	Strategy 1, 3
EMA filter	100-period, 1-min	Strategy 2
Exit persistence	$N = 4$ bars (20 min)	Strategy 3
Position sizing	Volatility-targeted	All versions
EOD liquidation	15:58 ET	All versions

Table 7. Strategy 4 vs. Strategies 2 and 3: targeted pairwise comparison.

Metric	S2	S3	S4
Total return (%)	227.00	252.00	**259.24**
Sharpe Ratio	0.94	1.012	**1.036**
Max drawdown (%)	8.7	9.3	**8.4**
Win rate (%)	40	40	**40**
Total orders	3,450	3,550	**3,388**

Observed results.

Interpretation. Strategy 4 achieves the best risk-adjusted and absolute performance of all five versions across the full 8-year horizon. Several features of the result merit detailed quantitative discussion.

Return and Sharpe. The total return of 259.24% marginally exceeds Strategy 3's 252% and substantially outperforms Strategy 2's 227%. More importantly, the Sharpe Ratio of 1.036 improves on both predecessors (0.94 and 1.012 respectively). The incremental Sharpe gain over Strategy 3 (+0.024) is modest in absolute terms but meaningful in context: Strategy 3 already filters out a large fraction of the baseline noise, leaving less room for further improvement from entry filtering alone.

Sortino Ratio and downside risk. The Sortino Ratio of 1.438 is particularly informative. While the Sharpe Ratio penalises all return volatility symmetrically, the Sortino Ratio penalises only downside deviations below the target return. A Sortino/Sharpe ratio of $1.438/1.036 \approx 1.39$ implies that a substantial fraction of Strategy 4's total return volatility is upside, consistent with the positive P&L skew expected of a trend-following strategy with asymmetric payoffs. This is comparable to Strategy 0's baseline ratio of $1.075/0.835 \approx 1.29$, confirming that the iterative refinements compound absolute performance while preserving and enhancing the favourable distributional character of the original framework.

Drawdown. At 8.4%, Strategy 4's maximum drawdown undercuts Strategy 3 (9.3%) and closely approaches Strategy 2's minimum (8.7%). Both represent a meaningful reduction from Strategy 0's 11.2%, which was itself already a constrained figure reflecting the inherently intraday nature of the strategy.

Expectancy and order efficiency. The positive Expectancy Ratio of 0.220 confirms that the strategy is structurally profitable in expectation on each trade, even with a 60% loss rate. This is the defining mathematical property of trend-following strategies: the average win (+0.74%) must substantially exceed the average loss (−0.36%) to generate a positive net return under a sub-50% win rate. Formally, the expected return per trade is:

$$E[\text{Return}] = w \cdot \bar{w} - (1-w) \cdot |\bar{l}| = 0.40 \times 0.74 - 0.60 \times 0.36 = 0.080\% > 0 \tag{12}$$

where $w = 0.40$ is the win rate. While the absolute expected return is 0.080% per trade, QuantConnect's dimensionless Expectancy Ratio normalises this to 0.220. Strategy 4's score of 0.220 improves on Strategy 0's baseline of 0.180, driven by a higher average win (0.74% vs. 0.68%) reflecting the selection of higher-quality breakout entries through EMA confirmation.

Probabilistic Sharpe Ratio. The PSR of 83.44% quantifies the probability that the true Sharpe Ratio exceeds a benchmark threshold (here, the Sharpe of a zero-alpha strategy), adjusting for the finite sample size and the observed skewness and kurtosis of the return distribution. A PSR exceeding 80% is the conventional threshold for statistical confidence in a strategy's edge; Strategy 4 clears this bar, providing formal support for the hypothesis that its Sharpe is not purely a sample artefact. This represents a material improvement over Strategy 0's PSR of 61.57%, which falls well short of the 80% threshold.

Pairwise comparison: Strategy 2 vs. Strategy 3 vs. Strategy 4. Table 7 isolates the three most competitive strategy variants. Strategy 2 achieves the lowest drawdown through entry filtering but sacrifices return and Sharpe relative to Strategy 3. Strategy 3 achieves the highest return among single-mechanism strategies through a noise-robust exit gate but carries marginally higher drawdown than Strategy 2. Strategy 4 synthesises the best properties of both: it matches Strategy 2's drawdown profile (8.4% vs. 8.7%), exceeds Strategy 3's total return (259% vs. 252%), and improves the Sharpe above both (1.036). The 3-month window results in Table 8 confirm this ordering is not period-specific: Strategy 4 achieves a Sharpe of 1.29 and a Sortino of 2.158 in the elevated-volatility Feb–Apr 2025 window, while generating fewer orders (114) than Strategy 3 (116) and matching Strategy 2's drawdown of 5.1%.

5. NEW BACKTESTING RESULTS AND CONCLUSIONS

5.1. Comparative Performance Summary

Fig. 2 shows the equity curves for all versions alongside the SPY benchmark over the full 8-year horizon; Fig. 3 isolates the February–April 2025 window to highlight the divergence during the high-volatility episode.

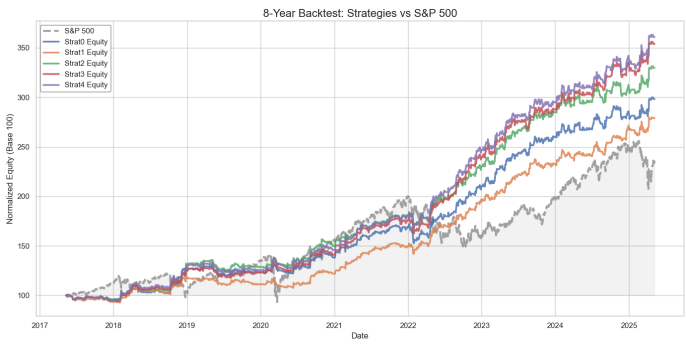

Figure 2. Cumulative equity curves for all five strategy versions and the SPY Buy-and-Hold benchmark, May 2017 – May 2025. Strategy 4 (EMA filter + temporal persistence) achieves the highest terminal wealth at 259% while maintaining a drawdown profile competitive with Strategy 2.

Table 8. Strategy evolution: comparative performance metrics — 8-year full period (May 2017 – May 2025) and recent 3-month window (Feb–Apr 2025), initial capital $100,000.

		8-Year Period				
Version	**Key Feature**	**Return (%)**	**Sharpe**	**Max DD (%)**	**Win Rate (%)**	**Orders**
Strategy 0	Base (30-min uniform)	196.00	0.835	11.2	40	3,594
Strategy 1	Decoupled freq. (5-min exit)	177.00	0.84	10.1	32	4,900
Strategy 2	EMA trend filter	227.00	0.94	8.7	40	3,450
Strategy 3	Temporal persistence	252.00	1.012	9.3	40	3,550
Strategy 4	**EMA filter + persistence**	**259.24**	**1.036**	**8.4**	**40**	**3,388**
		Recent 3-Month Window (Feb–Apr 2025)				
Version	**Key Feature**	**Return (%)**	**Sharpe**	**Max DD (%)**	**Win Rate (%)**	**Orders**
Strategy 0	Base (30-min uniform)	4.70	0.826	5.4	39	122
Strategy 1	Decoupled freq. (5-min exit)	4.79	0.91	5.5	28	158
Strategy 2	EMA trend filter	5.79	1.14	5.1	40	116
Strategy 3	Temporal persistence	6.36	1.30	5.1	41	116
Strategy 4	**EMA filter + persistence**	**6.30**	**1.29**	**5.1**	**40**	**114**

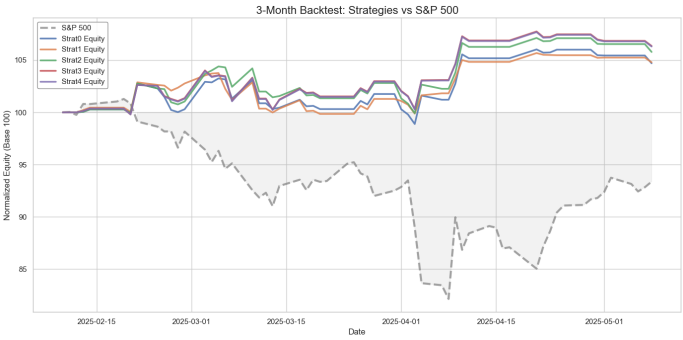

Figure 3. Cumulative equity curves, February–April 2025. While the benchmark sustains a drawdown exceeding 19%, all five active strategies preserve capital and generate positive returns. Strategy 4 leads on risk-adjusted performance (Sharpe 1.29, Sortino 2.158, max DD 5.1%).

The consolidated performance metrics for all five strategy versions are presented in Table 8.

5.2. Evolution Analysis: The Momentum Paradox

The progression across the five strategies illustrates a fundamental tension in high-frequency trend-following: *aggressive risk management can destroy the very returns it seeks to protect*. We label this the **Momentum Paradox**.

Strategy 0 establishes the baseline: a clean, interpretable volatility-breakout model with a Sharpe ratio of 0.835, a 196% total return, and a maximum drawdown of 11.2%. Strategy 1's naive fix—increase exit monitoring frequency to 5 minutes—achieves only a marginal drawdown improvement (11.2% → 10.1%) while degrading total return (196% → 177%), Sharpe (0.835 → 0.84, a near wash), and win rate (40% → 32%), confirming that adding exit complexity without a noise filter suppresses trade quality. Strategy 2 takes a conceptually orthogonal approach: by filtering entries rather than exits, it reduces the trade set and recovers the win rate to 40%, with the best absolute drawdown of 8.7%. Strategy 3 synthesises frequency and persistence: fast exit monitoring wrapped in a temporal persistence gate eliminates noise-driven false exits, achieving simultaneously higher profitability (252%) and tighter risk control (Sharpe 1.012, drawdown 9.3%) than any predecessor. Strategy 4 closes the loop by combining the entry-quality filter of Strategy 2 with the exit-robustness mechanism of Strategy 3, achieving 259% total return, a Sharpe of 1.036, a Sortino of 1.438, and a drawdown of 8.4%—the Pareto-dominant outcome across all five versions.

5.3. Regime Sensitivity

The 8-year backtest spans three materially distinct market regimes. During the low-volatility steady uptrend of 2017–2019, the dynamic leverage scales to 1.5× and the EMA filter is reliable. During the Covid crash and recovery of 2020–2021, the leverage scalar de-risks in March 2020 while the persistence gate prevents premature exits during the noisy recovery. During the rate cycle and episodic volatility of 2022–2025, trend days are more frequent and the recent absolute outperformance versus the benchmark is concentrated. Strategy 4, combining entry filtering with dynamic leverage and the persistence gate, is therefore the most broadly robust variant across all three regimes.

5.4. Performance Dashboard

Fig. 4 provides a consolidated visual comparison of all five strategy versions across five key performance dimensions. Strategy 4 (purple) leads on return and order efficiency, Strategy 3 (orange) leads on Sharpe in isolation, and Strategy 2 (green) achieves the tightest single-metric drawdown control.

5.5. Day Of The Week Effect

As documented by Zarattini et al. (2024) and the broader academic community (e.g., Dubois and Louvet, 1996), equity markets exhibit structural anomalies linked to the day of the week. In further analysis of our trades backtest, we confirm that this behavior is a robust feature of market microstructure rather than a mislead caused by overfitting. As detailed in Table 9, breaking down the generated trades by weekday exposes a remarkably consistent behavioral pattern across all strategy iterations, systematically following a strict Net Profit hierarchy:
Friday > Wednesday > Thursday > Tuesday > Monday.

Monday consistently emerges as the most critical trading session. It records both the lowest volume of executed trades and the lowest average profit per trade across all models. This underperformance occurs because early-week markets typically lack clear direction, exhibiting choppy, mean-reverting behavior that frequently triggers false breakouts. On the contrary, profitability surges from Wednesday onward. Mid-week sessions generate the sustained, directional trends our momentum algorithm requires.

Ultimately, the empirical results highlight **Friday** as the absolute performance peak. The magnitude of this outperformance is substantial: across the various iterations, Friday's average profit per trade is consistently five to ten times higher than Monday's.
In the S4 model, Friday alone generates a Net Profit of $99,357, averaging over $289.67 per trade.

Table 9. Day of the Week Effect: Detailed Strategy Comparison

	Total Trades					Win Rate (%)					Avg Profit ($)					Net Profit ($)				
Day	S0	S1	S2	S3	S4	S0	S1	S2	S3	S4	S0	S1	S2	S3	S4	S0	S1	S2	S3	S4
Monday	306	421	294	299	289	36.93	29.69	36.39	35.45	34.95	22.11	39.05	23.93	38.15	33.22	6,765	16,442	7,035	11,407	9,601
Tuesday	369	511	347	371	340	39.02	31.31	39.19	42.32	42.65	57.89	46.90	97.68	107.70	124.01	21,361	23,964	33,895	39,954	42,162
Wednesday	342	440	331	334	322	40.35	33.86	40.48	37.43	37.27	146.01	128.96	165.08	196.03	223.02	49,936	56,744	54,642	65,474	71,811
Thursday	413	586	401	417	400	39.47	29.35	39.65	39.57	39.0	114.96	41.19	122.52	127.29	130.36	47,477	24,139	49,132	53,080	51,244
Friday	367	492	352	354	343	47.41	40.04	48.30	46.33	46.36	234.59	154.03	279.56	278.38	289.67	86,093	75,781	98,406	98,545	99,357

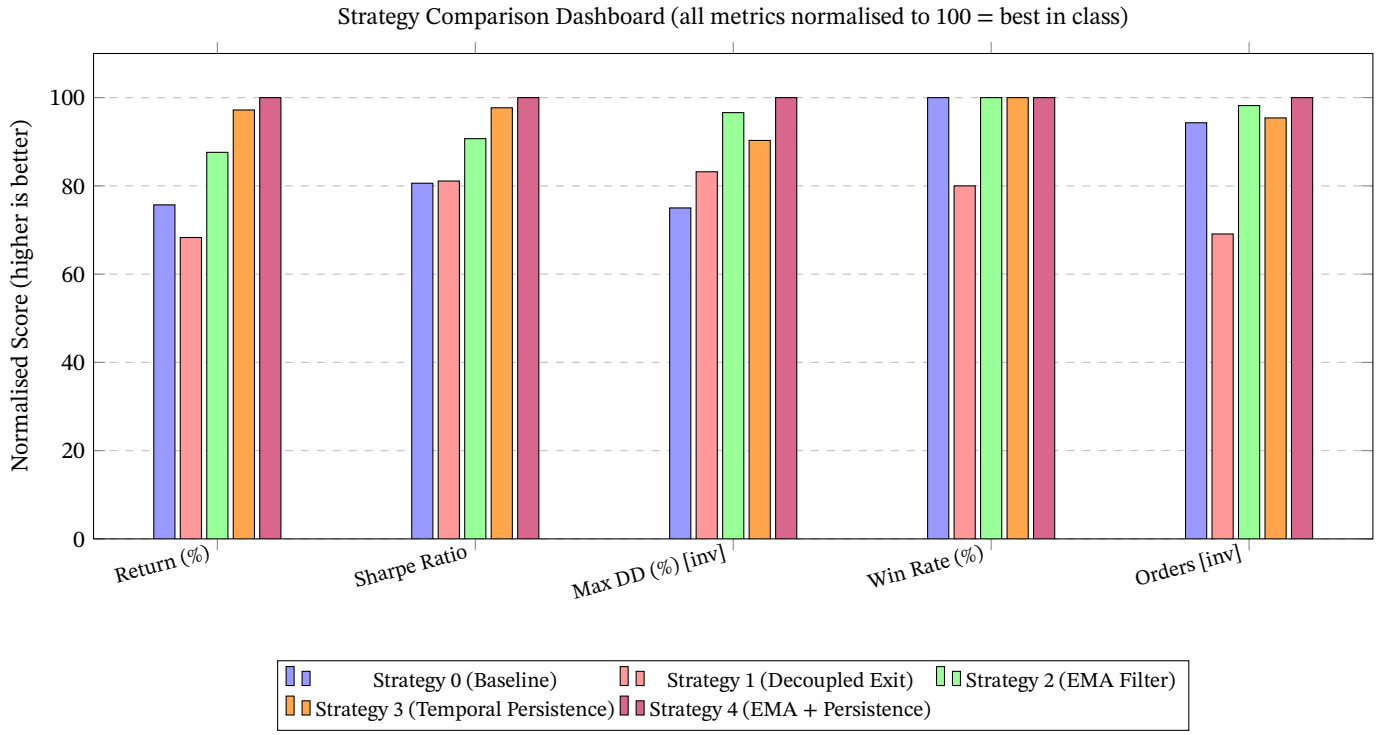

Figure 4. Comparative performance dashboard for all five strategy versions. Each metric is normalised so that the best-performing strategy scores 100. For maximum drawdown and total orders, scores are inverted (lower is better). Strategy 4 (purple) is the Pareto-dominant variant, leading or tying on all five metrics.

6. AVENUES FOR FURTHER DEVELOPMENT

The current strategy, while performant, rests on several simplifying assumptions that represent tractable research questions for future iterations.

6.1. Regime-Conditional Parameter Switching

The current framework uses fixed parameters ($N = 4$ persistence bars, 100-period EMA, 14-day lookback) across all market regimes. A natural extension conditions strategy parameters on a real-time regime classifier. Zarattini et al. (2024) document a monotonically increasing relationship between the VIX level at market open and the strategy's Sharpe Ratio up to approximately VIX 40, providing empirical support for regime-dependent parameter adaptation: a smaller persistence threshold N during elevated-VIX regimes (where intraday trends are more persistent) and a more conservative threshold during low-VIX regimes (where mean-reversion dominates). A Hidden Markov Model fitted to realised volatility and intraday return skewness could provide a probabilistic regime label that adjusts entry signals and persistence thresholds dynamically. An intraday breadth filter—the percentage of S&P 500 constituents trading above their intraday VWAP—could further reduce false signals on days where SPY's move is driven by a small number of high-weight constituents.

6.2. Volatility Multiplier Calibration

Zarattini et al. (2024) introduce a Volatility Multiplier (VM) parameter that scales the width of the noise bands. Their analysis shows that the best risk-adjusted returns are achieved at VM $\approx$ 1.5, whereas the default VM $=$ 1 used in our implementation maximises total return at the expense of a slightly lower Sharpe ratio. A systematic optimisation of VM within our QuantConnect framework, followed by walk-forward out-of-sample validation, represents a low-complexity, high-potential improvement.

6.3. Asymmetric Long/Short Treatment

The current strategy treats long and short signals symmetrically. An asymmetric persistence threshold—more conservative for long entries during downtrending markets, more aggressive for short entries during those same periods—could refine the asymmetry without sacrificing the short-side edge identified in the source strategy.

6.4. Multi-Asset Extension

The strategy is currently single-asset (SPY). Zarattini et al. demonstrate that the same volatility-breakout framework applied across 33 futures markets produces a diversified portfolio Sharpe of 1.65. Natural extensions within QuantConnect include sector ETFs (XLK, XLF, XLE) and E-mini futures (ES, NQ), which offer near-zero bid-ask spread and 23-hour liquidity. When running multiple instruments simultaneously, the leverage formula should be extended to account for intraday cross-asset correlation.

6.5. Optimisation of the Persistence Parameter

The choice of $N = 4$ exit confirmation bars is motivated by qualitative reasoning about intraday noise timescales. A rigorous in-sample opti-

misation followed by walk-forward out-of-sample validation would provide a statistically defensible choice. The optimisation surface should be examined for flatness around $N = 4$: a sharp peak would indicate overfitting risk, while a flat plateau would provide confidence in the robustness of the selection.

6.6. Order Execution Modelling

The current backtest uses QuantConnect's flat fee model ($0.005/share). A more realistic simulation would incorporate market impact using a framework such as the I-Star Market Impact Model of Almgren and Chriss (2001). Using limit orders for exits rather than market orders can convert the strategy from a pure taker to a partial maker, potentially reducing effective transaction costs by 50% or more on exit legs.

6.7. Statistical Robustness Tests

Before live deployment, the strategy should pass a battery of robustness tests: a Monte Carlo permutation test on Sharpe ratios under zero serial dependence, a parameter sensitivity analysis varying each free parameter by ±20%, subperiod analysis on rolling 2-year windows to detect strategy decay, and re-running with 2× and 5× baseline transaction costs to establish the break-even cost level.

7. CODE SOURCE

The computational framework and source code described in this study were developed by the authors within **BlackSwanQuants**. The complete repository, including implementation scripts and documentation, is available at:
https://github.com/blackswan-quants/intraday-momentum

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